

DBDOME -- LDAP / Active Directory Login Configuration Guide

DBDOME can authenticate its users against your corporate directory (Microsoft Active Directory or any LDAP v3 server). Users then sign in to the DBDOME UI with their own domain credentials, are created automatically on first login, and receive their DBDOME role from the Active Directory groups they belong to.

Everything is configured from one page -- no files to edit, no server access needed. The built-in local users (dbdome, admin) **always keep working**, so a wrong LDAP configuration can never lock you out.

1. What to collect before you start

Ask your Active Directory / IT team for the following:

Item	Example	Notes
Directory server host	dc1.corp.local	Name or IP of a domain controller
Port + encryption	636 (LDAPS)	LDAPS on 636 recommended; StartTLS or plain use 389
CA certificate (PEM)	corp-root-ca.crt	Needed when the directory uses an internal CA
Service account	CN=svc_dbdome,OU=Service Accounts,DC=corp,DC=local	A read-only account used only to search the directory; a regular low-privilege user is fine
Service account password	--	Stored encrypted by DBDOME
Users base DN	OU=Users,DC=corp,DC=local	Where your user objects live; several DNs are allowed
AD groups for roles	CN=DBDOME-Admins,OU=Groups,DC=corp,DC=local	One group per DBDOME role you want to hand out
A test user + password	jdoe	Any normal account, used once to verify the setup

2. Open the configuration page

From the DBDOME UI: open the **Configuration** dashboard and click **LDAP Login -> LDAP / Active Directory** in the *Configuration Tools* panel (next to the SMTP and TLS-certificate tools). Or browse directly to:

`https://<dbdome-host>:8080/ldap_settings`

The top of the page shows the current state: whether LDAP is active, the DBDOME Grafana service status, and the configuration files in use.

3. Server section

Field	What to enter
LDAP host	The domain controller name or IP (dc1.corp.local)
Port	636 for LDAPS, 389 for StartTLS or unencrypted
Encryption	LDAPS (recommended), StartTLS, or None
Skip TLS certificate verification	Leave no . Use yes only temporarily while troubleshooting certificate issues
CA certificate	Paste the PEM text (-----BEGIN CERTIFICATE----- ...) of your internal root CA, if the directory certificate is not publicly trusted

4. Service account section

Field	What to enter
Bind DN / UPN	The service account, as a full DN (CN=svc_dbdome,OU=Service Accounts,DC=corp,DC=local) or UPN (svc_dbdome@corp.local)
Bind password	The account password. Shown as set once saved; leave blank on later edits to keep the stored one

The password is stored **encrypted** on the DBDOME host (the encryption key is kept outside the database, so DB backups never contain a usable secret).

5. User lookup section

Field	What to enter
User search base DN(s)	One per line, e.g. OU=Users,DC=corp,DC=local. Users anywhere under these DNs can log in
Search filter	(sAMAccountName=%s) for Active Directory (the default). %s is replaced by what the user types at login. For non-AD directories use e.g. (uid=%s)
Group search base DN(s)	Optional; only needed when groups live outside the user search bases

The attribute mapping (username = sAMAccountName, email = mail, groups =memberOf) fits Active Directory out of the box.

6. Group-to-role mappings

Each row maps one AD group to a DBDOME role. **The first matching row wins**, so put the most privileged groups on top.

DBDOME role	What it allows
Viewer	See all dashboards and data, read-only
Editor	Viewer + create and edit dashboards
Admin	Editor + manage users, data sources and settings

Rules:

- Group DNs must be the **full DN**: CN=DBDOME-Admins,OU=Groups,DC=corp,DC=local
- A user in none of the mapped groups is **denied login**
- A row with group DN * matches everyone -- add one last * -> Viewer row if every authenticated domain user should be allowed in
- *Grafana admin* additionally grants server administration of the UI -- reserve it for the DBA team's group

7. Save, Test, Apply

1. **Save** -- stores the settings. Nothing is activated yet.
2. **Test** -- enter a test username (and optionally their password) first. The test runs against your directory **without touching the live login**: - verifies the service-account bind, - finds the test user under your base DNs, - lists the user's groups and shows **which role they would receive**, - if a password was entered, verifies it (a real end-to-end login check). Fix and re-test until the result looks right.
3. **Enable LDAP login** -- set the dropdown to *enabled*.
4. **Apply & restart Grafana** -- writes the configuration and restarts the DBDOME UI service. Active browser sessions see a ~10 second interruption.

8. Verify

Open the DBDOME UI login page and sign in as the test user with their domain credentials. First login creates the account automatically with the mapped role. The /ldap_settings page should now show **LDAP in Grafana: ACTIVE**.

9. Disabling LDAP

Set *Enable LDAP login* to **disabled** and click *Apply & restart Grafana*. Domain logins stop; local users continue to work throughout.

10. Troubleshooting

Symptom	Likely cause / fix
Test: <i>service-account bind failed</i>	Wrong bind DN or password; account locked/expired; try the UPN form (svc_dbdome@corp.local)
Test: <i>user not found</i>	User is outside the base DN(s); wrong search filter for your directory type
Test: TLS / certificate error	Directory cert issued by an internal CA -- paste the CA PEM; check host name matches the certificate; as a diagnostic only, set skip-verify to yes
Test OK but login denied in the UI	User is in no mapped group -- add the group or a * catch-all row
Wrong role assigned	Mapping order -- first match wins; check the user's groups in the Test output
Apply fails / service does not come back	See the service state on the page. The previous configuration is kept as a .bak - ldap file next to Grafana's ini; local logins still work
Password check FAILED for the test user	The user's password is wrong, or the account is disabled / must change password at next logon

11. Security notes

- The service (bind) account only ever **reads** the directory -- give it no other privileges.
- The bind password is stored encrypted; the rendered configuration file on the host is readable by administrators and the system service only.
- Prefer LDAPS (or StartTLS) so credentials never cross the network in clear.
- Local emergency access is deliberately retained: keep the local admin password in your password safe.

12. Reference (for administrators)

Item	Value
Configuration page	https://<dbdome-host>:8080/ldap_settings
Settings storage	config.ldap_settings (single row, password enc:v1: encrypted)
Rendered files	<conf>/ldap.toml, <conf>/ldap_ca.crt, [auth.ldap] in custom.ini (backup: custom.ini.bak-ldap)
Default conf dir	C:\ProgramData\DBDOME\conf (Linux: /etc/grafana)
Service restarted on Apply	DBDOME_Grafana (Linux: grafana-server)
Overrides (env vars)	GRAFANA_CONF, GRAFANA_INI, GRAFANA_SERVICE
API	/api/ldap/status · /api/ldap/save · /api/ldap/test · /api/ldap/apply